Literature Review

Frederik Appel Vardinghus-Nielsen

January 22, 2026

1 Introduction

Lubrication of ball bearings is essential for reducing friction, wear and overheating [41] [17]. Inadequate or excessive lubrication can lead to increased wear, higher operating temperatures, and ultimately premature failure of the bearing. Improper lubrication is estimated to be the cause of between 30% to 80% of ball bearing failures but the lubrication condition monitoring (LCM) has not received the same attention as the well documented vibration analysis used for both fault detection and remaining useful life (RUL) estimation [20] [9] [45] [33].

LCM of ball bearings covers the problems of estimating the film thickness, viscosity, degradation, and contamination from wear debris and environmental factors of the lubricant as well as rolling friction resulting from the conditions in the bearing. This in general is related to the ability of the lubricant to reduction friction and wear of the ball bearing.

This review aims to provide an overview of the current state of research in the field of online LCM for ball bearings. The review focuses on theoretical models of lubrication in ball bearings, sensing methods, signal processing techniques, and their applications in detecting lubrication degradation and film thickness changes.

1.1 Monitoring Flow

Online LCM relies upon information gathering during operation of ball bearings and consists of three or four distinct processes:

1. Sensing methods: observation and data collection from a ball bearing in operation through sensors.

2. Feature extraction: manipulation and analysis of the collected data through signal processing and machine learning.

3. Lubrication condition assessment: classification or regression of the processed data in order to infer lubrication conditions in the bearing.

4. Maintenance decision: a concrete relubrication strategy based upon the lubrication condition assessment. This includes when and how much to lubricate in order to maintain or restablish preferred lubrication conditions.

The steps in LCM related to the maintenance of a ball bearing are visualized in Figure 1. LCM is concerned mainly with inference on the lubrication conditions which stricly speaking does not include the maintenance decisions. The maintenance decision can, however, be included for a complete pipeline.

1.2 Existing Reviews

A number of reviews on lubrication condition monitoring in ball bearings have been conducted:

- [41] reviews the use of lubrication condition monitoring for maintenance decision support. The authors mainly cover direct analysis of used lubricant resulting in variables characterizing the physical and chemical state of the lubricant as well as the mathematical models used for maintenance decision support. The sensing methods covered are invasive.

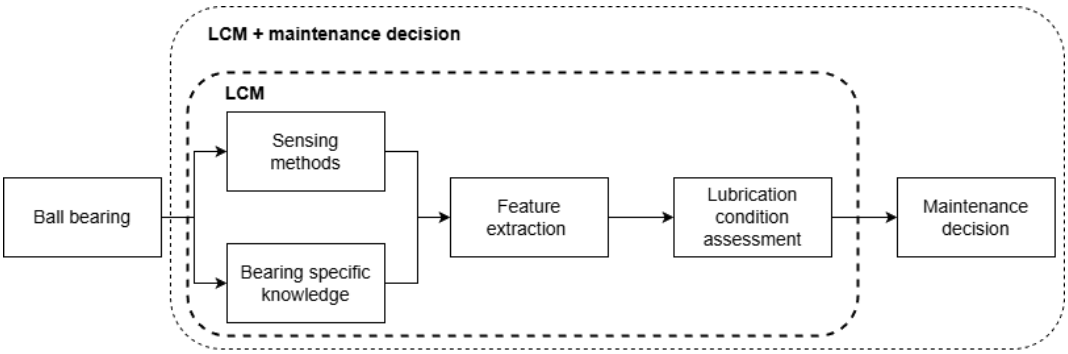

Figure 1: Steps in the process of maintenance of a running ball bearing with online LCM.

- [36] reviews the estimation of lubrication film thickness in bearings through the use of optical, electrical and ultrasonic sensing methods along with the physics models used for interpretation of the signals. It is concluded that the optical method is highly sensitive to film thickness but requires either unobstructed observation paths or alterations of the bearing structures. The electrical method requires highly sensitive sensors and the ultrasonic method is highly dependent on the bearing material and operating environment. All methods are active or invasive.
- [44] reviews bearing condition monitoring methods which includes detection of wear debris in the lubricant and lubricant film thickness. The lubricant analysis methods include offline lubricant laboratory analysis, ion-selective sensors mounted in the lubricant and magnetic methods relying on ferromagnetic or inductive sensors to detect debris content. All sensing methods are active or invasive.
- [14] reviews calibration methods for ultrasonic reflectometry. The methods are active and rely on ultrasonic waves induced into the bearing.
- [3] reviews the electrical capacitance methods which treats the bearing as a capacitor with the rolling elements and raceways acting as plates. The change in capacitance during operation can then be attributed to a change in the lubricant film between the plates.
- [1] similarly reviews electrical impedance based methods which seek to estimate the film thickness through changes in the impedance of the ball bearing system.
- [9] also reviews film thickness estimation based upon electrical capacitance models which rely on active sensors.
- [51] reviews online sensors for measuring lubricant properties but does not directly concern bearings. The authors mention sensors using magnetic induction, electrical capacitance, ultrasnoic emission and optical properties. All of the sensors are active and/or invasive.
- [38] reviews methods for monitoring debris content of lubricants which is related to wear of rotating machinery. The authors classify sensing methods into magnetic, electrical, optical and acoustic which are all active methods with the exeption of a ferromagnetic method.
- [11] reviews the ultrasonic reflection technology for estimating film thickness in lubricants. The review does not specifically concern bearings and the sensing mehods are active.
- [50] reviews lubrication oil monitoring methods and classifies them into electrical (magnetic), physical, chemical and opticel methods. The methods are all either active or invasive. Ball bearings are not specifically mentioned.

1.3 Existing Industrial Applications

Ultrasound is currently being used in the industry for passive, non-invasive, online LCM, e.g. through the Ultraprobe from UE Systems. This methods relies on a simple threshold model which establishes a reference level of ultrasound when the bearing is adequately lubricated and instructs to relubricate when the ultrasound energy level rises above a certain threshold set relative to the reference.

2 Literature Search Process

The literature search was conducted using the databases and search engines of Google Scholar. ScienceDirect, ResearchGate, Scopus as well as Connected Papers.

The search terms used included combinations of the following keywords: *lubrication*, *condition*, *monitoring*, *film*, *thickness*, *degradation*, *signal*, *processing*, *vibration*, *ultrasound*, *sound*, *acoustic*, *emission*, *surface*, *acoustic*, *wave*, *electrical*, *capacitance*, *impedance*, *magnetic*, *optical*, *infrared*, *signal*, *processing*, *ball*, *roller*, *rolling*, *bearing*, *oil*, *grease*, *online*.

3 Inclusion and Exclusion Criteria

A total of 131 sources were initially identified through the literature search process. Of the 131 sources, 48 were identified to treat the topic of lubrication condition monitoring for ball bearings. 33 of these treated monitoring with either active/inductive or invasive sensing methods while 20 treated passive and non-invasive sensing methods. These are used as the basis for this literature review.

4 Theoretical Background

Multiple theoretical models of lubrication exist which seek to estimate the lubrication film thickness in running ball bearings and the sufficiency thereof under specific operating conditions and with different lubricants. However, a complete understanding and model of the physico-chemical properties of lubricants and the lubricant film formation in ball bearings has yet to be formulated [3] [9] [24].

The following subsections present some of the most widely used models for estimating lubrication film thickness in ball bearings as well as a commonly used classification of lubrication regimes.

4.1 Lubrication Regimes

Lubrication films formed in ball bearings under operation are generally classifed into three different regimes which depend on the type of physical contact between balls and raceways and how the bearing load is distributed. The lubrication regimes are as follows:

- Boundary: the entire load is carried by surface to surface contact (ball to raceway). This regime provides the highest friction which stems solely from the sliding of the surfaces against each other.
- Mixed: the load carrying is shared between a thin lubricant film and surface asperities in the balls and raceway. This reduces friction and wear compared to boundary.
- Elastohydrodynamic: all load is carried by a lubrication film between the balls and raceway. The high pressure building up between the balls and raceway cause an elastic deformatZion of the solid materials.

4.2 Hamrock-Dowson Model

The Hamrock-Dowson model is an empirical model used for calculating the film thickness in ball bearings and is the most common model in use. It is given by

$$\frac{h_m}{R_x^{'}} = 3.63U^{0.68}G^{0.49}W^{-0.073}(1 - e^{-0.68k}) \tag{1}$$

where

- $R_x^{'}$ and $R_y^{'}$ are the effective radii in the direction parallel and transverse to the lubricant entrainment, respectively,
- $U = \eta_0 u_e/(E^{'}R_x^{'})$ is a dimensionless grouping of speed parameters, where u_e is the lubricant entrainment speed and $E^{'}$ is the effective elastic modulus of the balls and raceway,

- $W = w_z/(E^{'}R_x^{'})$ is a dimensionless grouping of load parameters, where w_z is the applied normal load,
- $G = \alpha^* E^{'}$ is a dimensionless grouping of material parameters, where α^* is the pressure-viscosity coefficient and
- k is an elliptic parameter, $k = 1.03(R_y^{'}/R_x^{'})^{2/\pi}$.

See [3], [15], [29] or [13] for further explaination of the variables and the physics motivated arguments for the equation.

4.3 Λ-Ratio

The Λ ratio is the ratio between the lubrication film thickness and the composite surface roughness of the bearing components and is used to assess the lubrication regime in a bearing, with higher values indicating better lubrication conditions [20] [16]. It is defined as

$$\Lambda = \frac{h_m}{\sqrt{R_{q1}^2 + R_{q2}^2}}, \tag{2}$$

where h_m is the minimum film thickness, and R_{q1} and R_{q2} are the root mean square roughness values of the two contacting surfaces. The rules of thumb for lubrication regimes based on the Λ ratio are as follows:

- $\Lambda < 1$: Boundary lubrication
- $1 \leq \Lambda \leq 3$: Mixed lubrication
- $3 < \Lambda$: EHL

In [15] it is noted that the Λ-ratio has shortcomings related to the elastic deformation of asperities in the EHL regime and that it it is unpredictably and may yield erroneous results. They therefore present an improved Λ^*-ratio given as

$$\Lambda^* = \frac{h^*}{z_0} = \frac{h_m + \delta_a}{z_0}, \tag{3}$$

where h_m is the minimum film thickness, δ_a is the elastic asperity deformation in an EHL regime and z_0 is the relaxed state height of the bearing. The Hamrock-Dowson model can be used for estimating h_m and methods for estimating δ_a are given in [15].

4.4 κ-Ratio

Tha κ-ratio is a model for assessing the adequacy of lubricant viscosity for a specific bearing under specific operating conditions. It is defined as

$$\kappa = \frac{\nu}{\nu_1}, \tag{4}$$

where ν is the modelled kinematic viscosity of the applied lubricant under specific operating conditions and ν_1 is the reference kinematic viscosity for the operating bearing, dependent on the bearing diameter and rotational speed as calculated from the ISO 281 standard [20]. The κ-ratio provides a non-invasive analytical method of estimating the lubrication condition from available operating parameters and lubricant properties. κ values are interpreted as follows:

- $\kappa \leq 0.1$: insufficient lubrication, bearing load is carried almost solely by contact between surface asperities.
- $0.1 < \kappa \leq 1$: insufficient lubrication, metallic surface to surface contact is present.
- $1 < \kappa \leq 4$: adequate lubrication.
- $4 < \kappa$: excessive viscosity resulting in higher friction.

The relationship between κ and Λ has been noted to be approximated as $\kappa = \Lambda^{1.3}$ and κ is noted to be time-invariant and ignore lubricant degradation and wear effects [20].

5 Sensing Methods

In this section an overview of the sensing methods used for LCM is givena and the commonly used signal processing methods or models are described for each of the sensing methods. Online sensing methods for collecting data which is informative on the lubrication condition in ball bearings can be categorized in multiple ways. In this section methods are categorized as either active or passive and invasive or non-invasive.

A sensing method is considered active if it induces, transmits or in another way supplies energy into the ball bearing system in order to allow a sensor to measure the resulting behaviour or response of the system. A sensing method is considered passive if it is not active, i.e. it solely measures signals which inherently stem from the bearing system under operation.

A sensing method is considered invasive if it requires unobstructed line of sight into the bearing, direct physical contact with the lubricant, mounting inside the bearing, altering the construction of the bearing or physically extracting samples of the lubricant. A sensing methods is considered non-invasive if it requires none of the above.

5.1 Invasive Methods

The below section briefly summarises non-invasive online sensing methods for lubrication condition monitoring in ball bearings. A categorization of the sensed physical quantities and signal processing methods is seen in Table 1.

5.1.1 Optical Methods

Optical methods employ light-based techniques to assess lubrication conditions. These methods include the use of fiber optic sensors, laser interferometry, or other optical measurement techniques to detect debris content which are then linked to lubricant viscosity [36] [29] [9]. The methods are invasive as they require direct line of sight to the lubricant and might need to be mounted in a way that alters the construction of the bearing.

Signal Processing
The following methods are used to derive lubrication conditions from optical measurements:

- Multi-target tracking: algorithms utilizing blob extraction and Kalman filtering track multiple debris particles to compute their velocity which is linked to bearing damage [27].
- Classification: convolutional neural networks are employed for particle classification distinguishing wear debris from air bubble interference based on image features [42].
- Statistical modeling: various statistical models including response surface methodology, ANOVA and regressions relate debris content to fluid viscosity [27].

5.2 Active Methods

The below section briefly summarises active sensing methods for lubrication condition monitoring. A categorization of the sensed physical quantities and signal processing methods is seen in Table 1.

5.2.1 Ultrasonic Detection Methods

Active ultrasonic methods utilize high-frequency sound waves to probe the lubrication film within a bearing. An ultrasonic wave is transmitted into the bearing and the modulation of this wave in the bearing contains information on the system. Ultrasonic methods can provide real-time measurements of film thickness and lubrication condition and operate on the principle that ultrasonic waves are partially reflected at the interface between the lubricant film and the bearing surfaces. The reflection coefficient is dependent on the film thickness and acoustic impedance of the materials involved. In conjunction with physical models of reflection and transmission the film thickness can be inferred from the measured ultrasonic signals [5] [4] [12].

Modelling

A reference signal is established by transmitting an ultrasonic signal into the bearing under known lubrication conditions and observing the modulation. The measured signal during operation is then compared to the reference signal to determine a reflection coefficient. The film thickness can then be calculated by using models of acoustic wave propagation and reflection [5] [12].

5.2.2 Electrical Methods

These methods consider the ball bearing and lubricant as parts of an electrical system where changes in the lubrication film thickness and condition affect the overall electrical properties of the system. Three different electrical concepts are used to estimate lubrication conditions: capacitance, impedance and resistance. All of the methods rely on an electrical current being induced into the bearing and then observing the modulation of signal as a consequence of the state of the bearing system. A reference signal is established under known conditions and signals under operation are then subsequently compared to the reference signal. [3] [1].

Modelling
Two main approaches are used to convert electrical measurements into film thickness:

- Physics based models of capacitance, impedance and resistance which establish expected relationships between lubrication conditions and the electrical signal used [3] [1] [35].
- Relative to a benchmark: Uses calibration tests with reference lubricants to derive an empirical relationship between the measured capacitance or output voltage and the calculated theoretical film thickness, e.g. using Hamrock-Dowson equations [3].

5.2.3 Surface Acoustic Wave

A surface acoustic wave (SAW) is an acoustic wave propagating along the surface of an elastic material and which decays quickly with depth in the material. Piezoelectric probes placed underneath the surface of the raceway actively excite SAWs that propagate along the surface of the raceway. As these waves interact with the lubricant film energy is transferred to the lubricant causing measurable wave attenuation and a time delay which can the be used to infer the lubrication condition [22]. The SAW can also be invasive if the sensors must be mounted inside the bearing.

Signal Processing and Modelling
SAW signals are used for inference about the lubrication conditions by either

- Signal processing: the time delay and amplitude attenuation is used to assess changes in lubrication conditions and the frequency content and moduation is correlated to the operating and lubrication conditions, e.g. estimated with the Hamrock-Dowson equations [25] [40].
- The physics based spring model is used to infer film thickness from SAWs reflected by the lubricant [40].

5.2.4 Magnetic Methods

Magnetic methods utilize the magnetic properties of certain lubricants, additives or wear debris to monitor lubrication conditions. By applying a magnetic field and measuring changes in magnetic flux or reluctance these methods can detect variations in lubricant film thickness and the presence of metallic debris [38].

Signal Processing
The magnitude of the output pulse signal is used to determine the size of the debris, and the polarity or phase of the signal is used to determine the material. Inductive sensors often use high-frequency AC excitation. The resulting modulated high-frequency AC signal must pass through a demodulation circuit and filter to eliminate harmonics and noise interference before the characteristic signal can be analyzed. Prior to feature extraction, the raw signal is typically subjected to zero-average processing to remove the DC component and enhance the accuracy of signal processing [38] [29] [37].

The symplectic geometry mode decomposition has furthermore been used to extract the debris signature of ball bearings [46].

5.3 Passive and Non-invasive Methods

The below section briefly summarises the different non-invasive or passive sensing methods for lubrication condition monitoring described in the literature. A categorization of the sensed physical quantities and signal processing methods is seen in Table 2.

5.3.1 Acoustic Emission

Aoustic emission (AE) are transient high frequency waves emitted by small deformations and stresses in solids. These are therefore linked to initial wear of materials. AE sensors detect these waves and operate in high-frequency bands often above 100 kHz. AE is highly sensitive to changes in friction, crack formation, plastic deformation on or beneath the surface of components and contamination in ball bearing making it suitable for monitoring lubrication conditions [16] [10] [44] [34].

Signal Processing
AE signals are processed using methods categorized into four main domains to diagnose bearing status and lubrication conditions:

- Time domain analysis: This involves statistical features such as root mean square, amplitude and kurtosis on the time series signal to trend overall signal level, impulsiveness, and energy content. AE parameters are also normalized to improve estimation accuracy across varying operating speeds [8] [22] [16] [34] [44].
- Frequency domain: This focuses on identifying characteristic frequencies and spectral components associated with bearing defects and lubrication issues using techniques like Fourier transform, envelope analysis, and spectral kurtosis [49] [22].
- Time-frequency domain analysis: This is used to analyze non-stationary signals arising from lubrication degradation, employing methods like short-time Fourier transform [22] [34] [44].
- Advanced signal processing, fusion and machine learning: Clustering based upon cross-correlation of signals is used to identify consistent AE source mechanisms such as crack and wear [34]. Minimum entropy deconvolution is used to enhance weak, random impulses caused by starved lubrication [45]. Lastly data fusion, convolutional neural networks and graph neural networks are used for detection of lubrication degradation [49] [22] [44].

5.3.2 Vibration

Rotating machinery produce vibrations which can be measured by accelerometers Vibrations are exacerbated by worsening lubrication conditions and bearing faults and the use of vibration analysis has been widely documented in literature and used in industry for bearing health monitoring. It has also been used for detecting lubrication faults but the method is limited in use because these often excite frequencies above 30 kHz which must be measured with other sensors [20] [49].

Signal Processing
Vibration signals are processed using methods categorized into four main domains to diagnose bearing status and lubrication conditions:

- Time domain analysis: this relies on estimating time-domain statistical measures, e.g. root mean square, kurtosis and crest factor, to quantify the overall signal level, variability, and impulsiveness. Impulsiveness metrics are particularly effective for detecting random mechanical impacts characteristic of starved lubrication or early faults [49] [20] [45].
- Frequency domain analysis: the vibration signal is transformed with the Fourier transform and spectral kurtosis is correlated to κ values to identify changes in lubrication conditions [20]. It is furthermore used in data fusion with AE signals for improved lubrication degradation detection [49].
- Signal enhancement and separation: the minimum entropy deconvolution is used to enhance weak, random impulses caused by events like lubricant starvation [45].

- Time-frequency domain analysis: techniques like the wavelet, Huang-Hilbert and short-time Fourier transforms are employed to analyze non-stationary signals arising from lubrication degradation [44].

5.3.3 Thermal Monitoring

Temperature sensors or infrared thermography detect heat generation caused by friction and wear inside the bearing, serving as an indicator for lubrication failure [49] [44].

Signal Processing
Temperature data is analyzed in the time domain focusing on trends and sudden increases in temperature that may indicate lubrication issues. [32].

5.3.4 Ultrasound >20 kHz

Ultrasound microphones are used to pick up high-frequency acoustic signals generated by frition and wear. In contrast to vibration sensors, ultrasound sensors can capture higher frequency components associated with lubrication degradation earlier in the lubrication degradation process [20]. Ultrasound based methods are currently being used in the industry to monitoring lubrication conditions. Examples include Sonotec, UE Systems and AccuTrak which produce ultrasonic probes used for such threshold based methods.

Signal Processing
Two different signal processing methods are currently used for processing ultrasound:

- Threshold methods: a baseline sound level under adequate conditions is established which subsequent measurements are then compared to. If the sound level rises a predetermined number of dBs, maintenance is adviced [26]. The industrial applications use this method.
- Time-domain analysis and regression: ultrasound has been processed using band pass filtering and time domain features combined with logistic models, LASSO regression and neural networks to predict values of κ, indicating correlation with lubrication conditions [20].

5.3.5 Sound 20 Hz - 20 kHz

Microphone arrays are used to capture audible sound emissions from the bearing. Changes in lubrication conditions increase the sound pressure levels and alter the frequency content of the emitted sound [23].

Signal Processing
Sound signals have been processed using frequency domain features which showed changes as lubrication levels change. It is suggested by the authors in [23] to use the features for classification of lubrication conditions through fuzzy logic or neural networks.

6 Invasive or Active Methods

Signal Processing Method	**Ultrasonic**	**Electrical**	**Inductive / Magnetic**	**Optical**	**Surface Acoustic Wave**
Time Domain	[36] [38] [26]	[38]	[38]	[38]	[22] [6] [25]
Frequency Domain		[38]			
Mode Decomposition			[46]		
Model-Based / Analytical	[36] [12] [43] [5] [51] [38] [11] [47]	[3] [1] [2] [7] [48] [30] [29] [9] [51] [35] [38] [18]	[51]	[2] [51] [27] [38] [18]	[6] [25]
ML / Deep Learning				[42]	
Sparse Sampling	[4]				

Table 1: Overview of active or invasive sensing methods for lubrication condition monitoring in rolling element bearings found in the literature and industry, categorized by signal processing methods used.

7 Non-invasive and Passive Methods

Signal Processing Method	**Vibration**	**Acoustic Emission**	**Temperature / Thermography**	**Ultrasound (>20 kHz)**	**Sound (20 Hz - 20 kHz)**
Time Domain	[20] [49] [39] [8] [21] [45] [19] [44]	[49] [31] [39] [8] [22] [16] [10] [28] [44]	[49] [32] [2] [44]	[20] [19] [26]	
Frequency Domain	[49] [21] [45] [19]	[49] [22] [28]	[49]	[19]	[23]
Time-Frequency / Decomposition	[21]	[22]			
Deconvolution	[45]				
Model-Based / Analytical			[32]		
Statistical Methods	[34]				
Machine Learning / DL	[49] [44]	[49] [22] [44]	[49] [44]		
Data Fusion	[49]	[49]	[49]		

Table 2: Overview of non-invasive and passive sensing methods for lubrication condition monitoring in rolling element bearings found in the literature and industry, categorized by signal processing methods used.

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